

Norway

SLEIPNER

Volve F

F-10

F-10

Plan: F10 Rev 5.1 280409

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F10 Rev 5.1 280409	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-10			
Well Position	+N/-S	-3.91 m	Northing:	6,478,559.62 m
	+E/-W	2.23 m	Easting:	435,052.25 m
Position Uncertainty	0.00 m		Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-10				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2008	05.04.2009	-2.02	71.61	50,402

Design	F10 Rev 5.1 280409				
Audit Notes:					
Version:	Phase:	PLAN	Tie On Depth:	1,381.60	
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)	
	145.90	-3.91	2.23	124.70	

Survey Tool Program	Date	21.02.2013			
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description	
150.00	260.00	26" Continuous wireline gyro to SB (F-10)	Keeper, cont	Gyro Tool from SDC	
266.90	683.00	26" GyroMWD survey (F-10)	Keeper, stat	Gyro Tool from SDC	
709.50	1,381.60	26" MWD survey (F-10)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	
1,381.60	5,413.67	F10 Rev 5.1 280409 (F-10)	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)	

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
1,381.60	27.47	120.39	1,324.51	-158.61	267.66	306.29	0.000	0.000	0.000	
1,430.00	27.95	118.95	1,367.36	-169.75	287.22	328.71	0.511	0.298	-0.893	
1,463.00	27.75	118.00	1,396.54	-177.10	300.77	344.03	0.443	-0.182	-0.864	
1,730.00	25.08	118.00	1,635.64	-232.85	405.63	461.99	0.300	-0.300	0.000	
1,780.00	28.41	118.00	1,680.28	-243.42	425.50	484.34	2.000	2.000	0.000	
2,028.80	45.00	126.29	1,879.05	-322.79	550.49	632.29	2.083	2.000	1.000	
2,765.53	45.00	126.29	2,400.00	-631.15	970.38	1,153.03	0.000	0.000	0.000	
3,264.97	78.33	119.00	2,633.96	-860.98	1,337.37	1,585.59	2.037	2.002	-0.438	

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Planned Survey									
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3,391.88	86.05	122.50	2,651.20	-925.24	1,445.30	1,710.91	2.000	1.824	0.828
4,178.33	86.05	122.50	2,705.41	-1,346.81	2,107.00	2,494.92	0.000	0.000	0.000
4,320.41	85.63	131.99	2,715.74	-1,432.48	2,219.68	2,636.32	2.000	-0.088	2.003
4,727.12	85.63	131.99	2,746.73	-1,703.78	2,521.09	3,038.57	0.000	0.000	0.000
5,037.19	55.00	126.80	2,850.00	-1,887.78	2,743.12	3,325.86	3.000	-2.964	-0.502
5,263.67	55.00	126.80	2,979.90	-1,998.91	2,891.67	3,511.25	0.000	0.000	0.000
5,413.67	55.00	126.80	3,065.94	-2,072.52	2,990.07	3,634.05	0.000	0.000	0.000

Design Targets									
Target Name	Dip Angle (°)	Dip Dir. (°)	TVD (m)	+N/-S (m)	+E/-W (m)	Northing (m)	Easting (m)	Latitude	Longitude
Theta Sør T#2.3 MM - hit/miss target - Shape - Point	0.00	0.00	2,922.90	-2,340.34	3,162.08	6,476,224.00	438,211.00	58° 25' 15.8283 N	1° 56' 32.0767 E
- plan misses target center by 348.97m at 5413.67m MD (3065.94 TVD, -2072.52 N, 2990.07 E)									
Theta Sør T#2.2 MM 13I - plan misses target center by 67.65m at 5263.67m MD (2979.90 TVD, -1998.91 N, 2891.67 E) - Point	0.00	0.00	2,971.90	-2,057.24	2,925.00	6,476,507.00	437,974.00	58° 25' 24.8560 N	1° 56' 17.1984 E
Theta Sør Prelim 1 1606 - plan misses target center by 90.82m at 5259.53m MD (2977.52 TVD, -1996.88 N, 2888.96 E) - Point	0.00	0.00	2,979.90	-1,923.19	2,942.00	6,476,641.00	437,991.00	58° 25' 29.1964 N	1° 56' 18.1156 E
Theta Sør T#2.1 MM 13I - plan misses target center by 0.44m at 5263.67m MD (2979.90 TVD, -1998.91 N, 2891.67 E) - Point	0.00	0.00	2,979.90	-1,999.22	2,891.99	6,476,565.00	437,941.00	58° 25' 26.7140 N	1° 56' 15.1085 E
E TS - plan misses target center by 514.79m at 3801.94m MD (2679.46 TVD, -1145.05 N, 1790.32 E) - Point	0.00	0.00	3,064.90	-843.82	1,950.66	6,477,720.00	437,000.00	58° 26' 3.5664 N	1° 55' 15.9815 E
Theta Sør DRAFT Tgt - plan misses target center by 437.09m at 3662.24m MD (2669.83 TVD, -1070.17 N, 1672.78 E) - Point	0.00	0.00	3,070.90	-1,199.94	1,557.24	6,477,364.00	436,606.72	58° 25' 51.8537 N	1° 54' 52.0946 E
F-X VS - plan misses target center by 2166.08m at 2762.47m MD (2397.83 TVD, -629.87 N, 968.63 E) - Point	0.00	0.00	3,114.90	-1,748.13	-742.28	6,476,816.00	434,308.00	58° 25' 32.9179 N	1° 52' 30.9848 E
E VS - plan misses target center by 1945.89m at 2592.08m MD (2277.35 TVD, -558.55 N, 871.52 E) - Point	0.00	0.00	3,133.90	-1,278.97	-720.27	6,477,285.00	434,330.00	58° 25' 48.0905 N	1° 52' 31.8572 E
Volve Sør Injector 1201C - plan misses target center by 1945.89m at 2592.08m MD (2277.35 TVD, -558.55 N, 871.52 E) - Point	0.00	0.00	3,133.90	-1,278.97	-720.27	6,477,285.00	434,330.00	58° 25' 48.0905 N	1° 52' 31.8572 E
E W2 - plan misses target center by 2355.45m at 2004.17m MD (1861.51 TVD, -313.80 N, 535.73 E) - Point	0.00	0.00	3,264.90	838.77	-964.36	6,479,402.00	434,086.00	58° 26' 56.3915 N	1° 52' 14.6281 E

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Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-10	North Reference:	Grid
Wellbore:	F-10	Survey Calculation Method:	Minimum Curvature
Design:	F10 Rev 5.1 280409	Database:	Production EDM P246N

Casing Points					
	Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
	202.00	202.00	30"	30.000	36.000
	1,388.92	1,331.00	20"	20.000	26.000
	2,607.14	2,288.00	13 3/8"	13.375	17.500
	3,435.00	2,654.17	9 5/8"	9.625	12.250

Checked By: _____	Approved By: _____	Date: _____
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